Synthesized Features

A persona framework for collections allocation

Group Retail Risk · Group AI, Data & Analytics · November 2026

SUMMARY

Synthesized features give allocation a view of the customer, built from data we already hold

TODAY

No model beyond 60 DPD. Cases are allocated by bucket and product, and late buckets hold little behaviour to model.

UC2 · SYNTHESIZED FEATURES

Signals from every data domain, condensed each month into four answers per account:

1. **How did they get here?**
2. **Can we reach them?**
3. **Will they pay?**
4. **Can they pay?**

WHAT IT SHOWS

54% of the money collected in the next 3 months came from the top 20% of card accounts ranked this way.

8× Within the same DPD bucket (91–120), some loan customers pay 8 times as often as others.

12% of 180+ accounts show a positive signal; one in five of them pays, against under 3% of the rest.

NEXT

By 20 Nov
UC2 synthesized features signed off; one may follow later

20 Nov – 11 Dec
UC1 transcript signals and UC2 combined, with action support

11 – 20 Dec
Pilot selection

From 4 Jan
Pilot

WHERE WE START

Allocation follows the bucket, not the customer

WHO WORKS THE CASE

In-house or agency, senior or junior

Decided by [bucket and balance: confirm], not by whether we can reach the customer or whether they can pay.

WHAT WE OFFER

Settlement, plan or push back

Follows [the bucket policy: confirm], not the customer's capacity or track record on this debt.

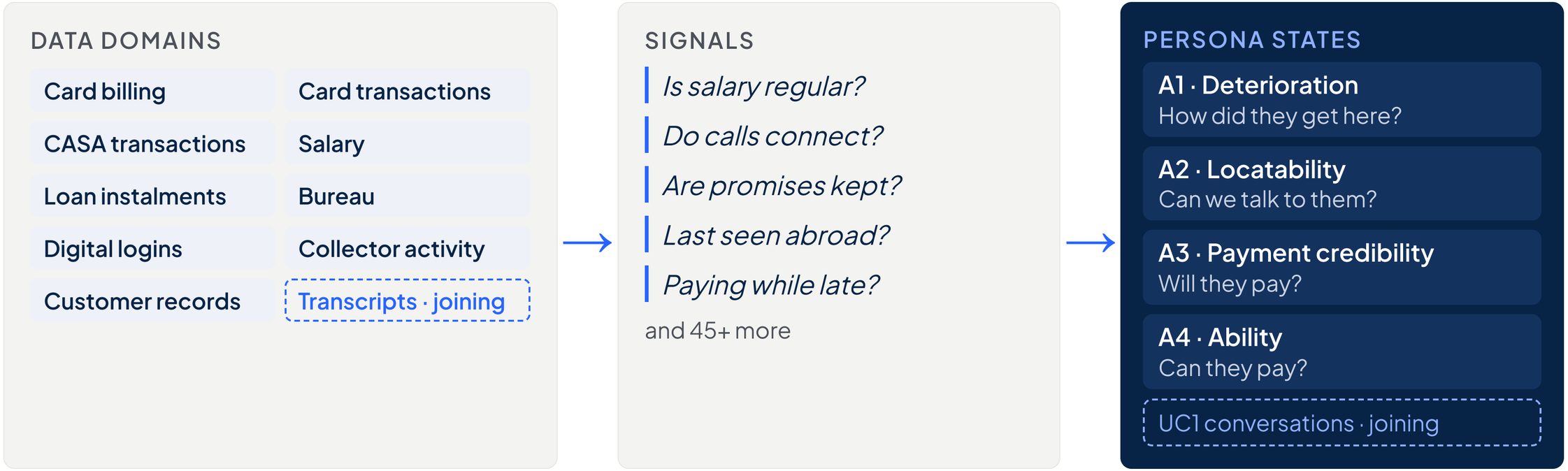
WHERE 180+ EFFORT GOES

374k accounts at 180+

Worked as one pool, though a small share still shows signs they could pay. Finding them is the 180+ goal.

SYNTHESIZED FEATURES

Everything we know about a customer, condensed into a few states



10 data domains · 50+ signals · 4 states per account, every month

61–180 DPD: personas that drive allocation

180+ DPD: find the customers with a positive signal

THE IDEA

Three questions before a case reaches a collector

1 How did they get here?

A1 · Deterioration

Curing now
stabilized

Cured or paid before
slow roller · repeat delinquent

Paid, then stopped suddenly
seasoned · early-life straight roller

Never paid
early payment default

→

2 Can we talk to them?

A2 · Locatability

The gate
where they were last seen · whether calls connect

Reachable on to step 3 →

Non-Reachable · calls fail

Skip · last seen abroad

→

3 What can we offer?

The nuance, for the reachable

A3 · Credibility · Will they pay?

High | Moderate | Low | Unknown

A4 · Ability · Can they pay? IN PROGRESS

High | Moderate | Low | Unknown

UC1 · Conversations · joining
intent and ability to pay, from call transcripts

INFORMS
Context and triage

DECIDES
Route: in-house, agency or tracing

DECIDES
Offer and agent

Decisions shown are examples · dashed = Unknown, or not yet live

HOW IT IS BUILT

Built to be explained and trusted

Rules you can trace

Every state comes from a written rule, and the record says which rule assigned it and why.

Unknown is a state

When evidence is missing we say so, with the reason, instead of guessing.

Point in time

Each month end uses only what was known that day: the view a strategy would have had.

Customer-level evidence

Salary, CASA, bounces and logins are read across all of a customer's accounts, not only the delinquent one.

PART 1

The four axes

Deterioration · Locatability · Payment credibility · Ability

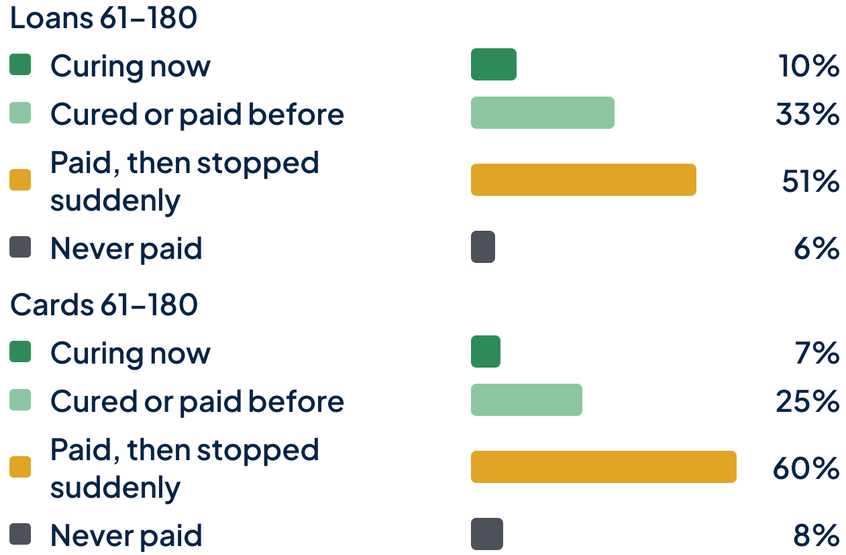
Customers curing now pay far more often than any other group

HOW A CUSTOMER GETS A LABEL

Read from the DPD history alone, so it covers every account. Asked in order; the first “yes” decides.

1. Nothing paid since the first months? → **Early payment default** — bad from the start
2. Holding at one level for months? → **Stabilized** — stopped falling
3. Back within a year of curing? → **Repeat delinquent** — been here before
4. Fell to 61+ fast, after a year of good history? → **Seasoned straight roller** — sudden stop
5. Fell to 61+ fast, in the first year? → **Early-life straight roller** — fell fast, early
6. Otherwise → **Slow roller** — fell slowly, paying on the way

AUGUST MIX



60% of card accounts curing now paid within 3 months, against 22% of those who cured before.

Half of loans (six in ten cards) paid for a while, then stopped all at once: an event, not a habit.

Production, 31 August 2026, 61–180 DPD · paid: meaningful payment within 3 months, back-test

A2 · LOCATABILITY STATES · DECIDED BY RULES

Only one in five card customers at 61–180 can be reached

HOW A CUSTOMER GETS A LABEL

Location Where were they last seen?

Logins and in-person card use show where they are; a travel purchase or collector note signals they have left. The latest evidence wins.

In Country | Skip | Unknown

Reachability Do our calls connect?

Recent call days all failed, no contact since → Non-Reachable. A recent contact → Reachable.

Reachable | Non-Reachable | Unknown

The gate: only Reachable customers go on to step 3.

AUGUST MIX

Location × reachability, 61–180

% of card accounts · loans below

	Reachable	Non-Reachable	Unknown
In Country	8.5% loans 10.2%	13.6% loans 14.2%	1.9% loans 2.7%
Skip	6.2% loans 8.1%	31.7% loans 28.5%	6.1% loans 6.8%
Unknown	6.3% loans 7.9%	22.3% loans 14.9%	3.4% loans 6.8%

1 in 3 look abroad and do not answer, the largest group on cards and loans alike: candidates for tracing or an agency.

3.7% of 180+ card accounts are reachable (loans 4.8%), a positive signal for finding those still worth working.

Within the same bucket, credible customers roll far less

HOW A CUSTOMER GETS A LABEL

Three sources earn points; we add them up

Promises kept
last 12 months

Kept +2 | Partly +1 | Broken −1

Paying while delinquent
regular = the monthly due paid in 2 of the last 3 months

Regular +2 | Some +1

Record before
clean = paid as billed before falling behind

Clean +1

High	Moderate	Low
4+ points	2–3 points	0–1 points

> An explicit refusal to pay → **one band down**

AUGUST MIX

	High	Moderate	Low
Cards 61–180	2.5%	19.4%	78.0%
Loans 61–180	4.9%	27.9%	67.2%
Cards 180+	0.1%	2.7%	97.2%
Loans 180+	0.3%	4.3%	95.4%

42% vs 90% of card customers at 91–120 DPD rolled further: High vs Low credibility.

59% of card customers made no promise to pay in the last 12 months: their credibility rests on payments alone.

Roll rates: back-test, 3-month outcomes compared within each DPD band

A4 · ABILITY IN PROGRESS

1 How they got here — 2 Can we talk to them — 3 What we can offer

Ability tells us what a customer can realistically be offered

HOW A CUSTOMER GETS A LABEL

Four evidence families; we count how many are strong*

Income
Regular salary, ours or bureau

Money in
Money arriving or held, not salary

Spending
The customer's own spending

Other products
Activity on their other products

High
2+ strong

Moderate
1 strong

Low
some evidence, none strong

Unknown
no evidence

> A cheque bounce or returned direct debit → **one band down**

AUGUST MIX

	High	Moderate	Low	Unknown
Cards 61–180	0.2%	14.4%	28.9%	56.6%
Loans 61–180	0.3%	8.9%	44.7%	46.1%
Cards 180+	0.0%	6.2%	6.4%	87.4%
Loans 180+	0.0%	3.5%	6.9%	89.5%

Needs: GRR's bureau salary feed

About a third of customers have no CASA with us, and half never had a salary credit here. Without a regular bureau salary feed they stay Unknown; with it, income becomes visible across the book.

57% of card customers at 61–180 show no evidence of their money yet.

6% of card accounts at 180+ show the means to pay: a positive signal.

* Strong = regular and above the floor in at least 2 of the last 3 months (floor: AED 3,000 or twice the amount due); at 180+: 3 of the last 6 months

PART 2

What the personas could support

A first look at allocation · Risk score · UC1

A FIRST LOOK

How an account gets a tier: by what its persona actually paid

1 Describe each account

One state on each of the four axes. Together they form its **persona**.

- **Stopped suddenly**
- **Skip · Non-Reachable**
- **Low credibility**
- **Unknown ability**

576 personas are possible; 272 occur on cards.

2 Measure what it paid

For each persona: every account in May 2026, everything it paid in June–August, averaged per account.

Example: 400 accounts paid AED 70,000 in total → **AED 175 per account**. 10% of them paid, AED 1,750 each on average.

Observed, not modelled: the average combines how often and how much.

3 Rank and cut

The 159 personas with at least 10 accounts (97% of cards), ranked by AED per account:

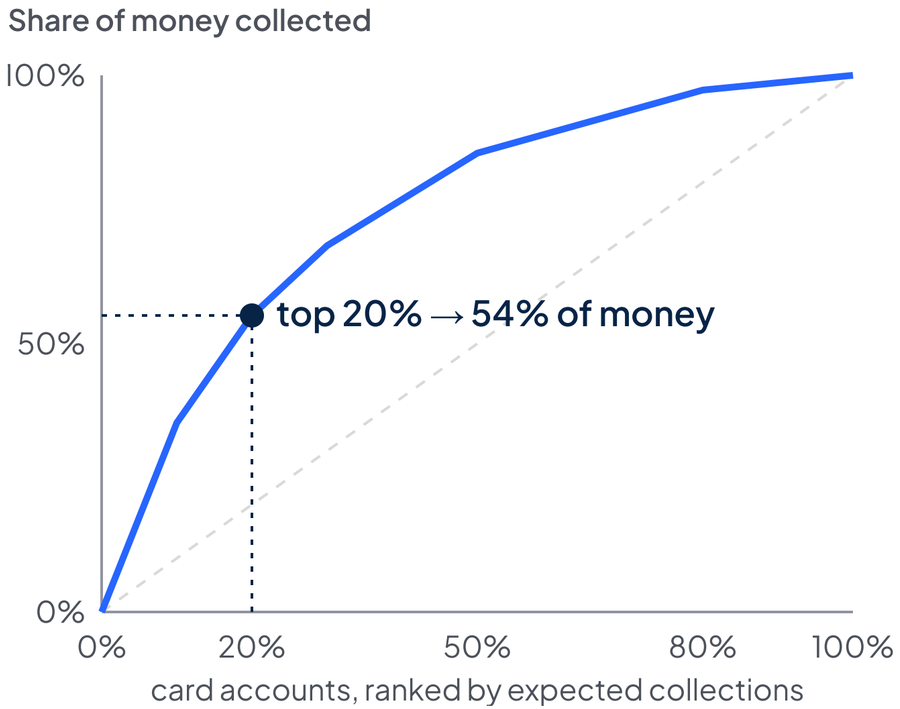
In-house	AED 1,250 and above
Agency	between
Non-workable	under AED 175

Illustrative cut-offs from the data; GRR would set them from the commission rate card and in-house cost.

Cards 61–180 · back-test May → Jun–Aug 2026; tiers hold on unseen months · personas under 10 accounts unassigned (3%)

A FIRST LOOK

The personas could give the policy's non-workable pool a definition



Tier	Accounts	Money collected	Recovery, 3 m	Call days
In-house	17%	48%	10.0%	11.7
Agency	54%	46%	2.7%	11.8
Non-workable	29%	6%	0.56%	10.9

Definition: accounts whose persona has returned under AED 175 in 3 months (about 0.5% of principal).

In practice: stopped paying suddenly (79%) or never paid (20%), and not reachable.

10.9 call days per account in 3 months still go to the non-workable pool, almost as many as in-house.

Cards 61–180 · table out-of-sample (5 folds, Jan–May 2026); call days: August · cuts AED 1,250 / 175 · loans: same order, thresholds less stable

RISK SCORE

Personas rank payers far better than the bucket; a model adds a little on top

How well each view ranks who pays (AUC) · 61–180

Ranking customers by	Cards	Loans
DPD bucket only (today)	0.65	0.62
Personas	**0.83**	**0.77**
Personas × DPD bucket	0.84	0.77
Simple model (axis states + bucket)	0.84	0.78

Out-of-sample: 5 folds by account, January–May 2026 cohorts. 0.5 = chance, 1.0 = perfect.

Personas beat the bucket in every test

+0.17 on cards, +0.14 on loans. The top 20% of card accounts by persona hold 54% of the money collected (bucket: 30%).

The bucket adds almost nothing within personas

+0.01 on cards, +0.00 on loans.

A model on the same inputs adds about **+0.02** over the personas. **Next:** test whether a full Retail 360 model adds more, then decide: with the personas or Phase 2.

Paid = meaningful payment within 3 months (the risk score's target) · strict and rolling time splits give the same picture

UC1

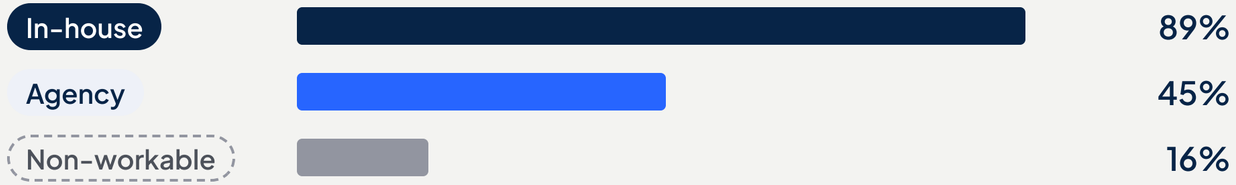
UC1 adds most where we already talk to customers

1. **How did they get here?**
   A1 · Deterioration
2. **Can we talk to them?**
   A2 · Locatability
3. **What can we offer?**
   A3 Credibility · A4 Ability · UC1 Conversations

What transcripts can add to the nuance

- Stated intent to pay
- Change in income or job
- Hardship reason
- Dispute
- Promise made on the call
- Third-party contact

Right-party contact in the last 3 months, cards 61–180



All cards 41% · loans 52%

Next: integrate UC1 into the final personas, then a strategy workshop with GRR on the decisions each supports.

APPENDIX

Signals, rules and governance

For the detailed discussion: where each signal comes from, the exact rules, and how the framework is controlled.

APPENDIX · ILLUSTRATIVE GROUPING, NOT A PROPOSAL

Family and reachability split payment rates from 3% to 68%

1 · How they got here	2 · Can we talk to them	3 · The nuance	Share	Paid in 3 m
Curing now · 7%	Reachable	Credible or able	4.1%	68%
		Neither	0.4%	52%
	Not reachable		2.1%	44%
Cured or paid before · 25%	Reachable	Credible or able	3.4%	40%
		Neither	2.7%	30%
	Not reachable		19.0%	16%
Paid, then stopped suddenly · 60%	Reachable	Credible or able	3.5%	34%
		Neither	6.1%	22%
	Not reachable		50.8%	8%
Never paid · 8%	Reachable	Any	0.6%	14%
	Not reachable		6.8%	3%

Cards 61–180 · paid = meaningful payment within 3 months (back-test)

APPENDIX · A FIRST LOOK

What each group could mean, for discussion

In-house

Curing now or cured before, reachable, credible or able

Convert through skilled collectors: settlement, restructure or deferral

Policy tools: CARES settlements, loan restructure, deferral

11% of cards · AED 40m

50% paid in 3 months

Agency

Stopped suddenly or cured before, not reachable

Place with agencies at 61–180, skip tracing where they look abroad

Policy: referral by product, DPD stage, contactability and strategy

54% of cards · AED 213m

17% paid in 3 months

Non-workable

Stopped suddenly or never paid, not reachable

Free collector time: refer to agencies or stop calling

Policy: non-workable pool to agencies (HOD approval), or no calling with dunning as for Dead Beat accounts

32% of cards · AED 138m

4% paid in 3 months

180+ The 12% with a positive signal: lift out of the pool for an in-house attempt. About one in five pays, against under 3% of the rest.

For discussion in the strategy workshop; not agreed with GRR · cards 61–180; AED = outstanding principal; paid = meaningful payment

APPENDIX · A FIRST LOOK

How a pilot could test it, if the strategy workshop takes it forward

Test in each tier

- **In-house:** skilled collectors and settlements vs business as usual
- **Agency:** placement at 61–180 vs in-house calling
- **Non-workable:** agency referral or dunning only vs business as usual (hold-out)
- Random split within each tier, 3 months

Measure

- Meaningful payment and roll rate
- Amount collected per account
- Recovery of outstanding principal
- Call days and cost per AED collected
- 180+: the flagged group vs the rest

Size and guardrails

- Cards: agency and non-workable fill both arms in one month
- In-house needs a 10-pt target; loans need 10 pts and pooling
- Same policy and compliance rules
- Monthly readout and a stop rule

Sample per arm (cards, 5-pt lift): agency 970, non-workable 375; in-house 388 at a 10-pt lift · two-sided α 0.05, 80% power

APPENDIX A

Signals and their sources

Axis	Signals	Sources
A1 Deterioration	Monthly DPD bucket path, restructures linked to their original loans	Way4 billing history, loan master
A2 Locatability	UAE vs foreign logins, in-person card use (POS, ATM) by country, departure notes, travel purchases, call outcomes, deceased flags	Digital login, Way4 card transactions, CASA card transactions, DCORE follow-ups, customer records
A3 Credibility	Promises and plans with their outcome, payments vs dues per cycle, record before delinquency, refusals, disputes	DCORE plans and follow-ups, card statements and payments, loan monthly dues and cash
A4 Ability	Salary per month (larger of ours and bureau), CASA balance and activity, external card payments, cheque / direct-debit returns, bureau bounces	Wealth-management salary and CASA features, AECB salary and bounce data, loan master (instalment)

Customer-level signals are read across every account of the customer (CIF). Windows end at the observation month end: 3 months at 61–180 DPD, 6 at 180+.

All sources are existing production datasets; no new data feed is needed to run the framework

APPENDIX B

The scoring rules at a glance

Axis	Evidence and points	State
A1 Deterioration	DPD path only, from January 2024 (reliable window)	First “yes” of six ordered questions (Appendix C)
A2 Locatability	Latest location evidence: foreign → Skip; UAE within 3 months (6 at 180+) → In Country; travel rule. Last 3 call days failed (6 at 180+) → Non-Reachable	Location × reachability (9 cells) + Deceased override
A3 Credibility	Promises: Kept ≥ 80% +2, Partly +1, Broken < 40% or last 3 broken –1. Payments: monthly due paid in 2 of the last 3 months +2, any payment +1. Record before: 80% paid as billed +1	High ≥ 4 · Moderate ≥ 2 · else Low; refusal –1 rung; no delinquency history → Unknown
A4 Ability	Four evidence families: income (salary, ours or bureau), money arriving or held, own spending, other products. In progress	High: 2+ strong · Moderate: 1 strong · Low: evidence, none strong; distress –1 band; no evidence → Unknown

Full specification and reference workbook per axis

APPENDIX C

How a journey state is assigned

#	Question	Yes when	State
0	Start of journey seen?	Oldest month already delinquent and not stabilized	Unknown
1	Early payment default?	≤ 3 months of history, rolled in, never paid since	EPD
2	Stabilized?	Pays about one month's due in 5 of the last 6 months (61–180 only)	Stabilized
3	Repeat delinquent?	Cured an earlier spell that reached 31–60 DPD, back within 12 months	Repeat delinquent
4	Straight roller?	Current to 61+ DPD in ≤ 3 months, no payment on the way	Seasoned (12+ months on book) or early-life
5	Otherwise	Paid something on the way down, kept rolling	Slow roller

Billing artefacts repaired first and counted · retail loans on the same tree, buckets shifted by one

APPENDIX D

Status, controls and sign-off

Axis	Status	Open with GRR
A1 Deterioration	In production; restructured loans linked	Auto-loan restructure recording
A2 Locatability	Accepted by GRR, 30 Oct 2026	Foreign-login (VPN) check; loan and 180+ result blocks
A3 Credibility	GRR deadline and thresholds decided; being applied	180+ back-test; 30-day deadline as an alternative
A4 Ability	In production, all four models	25% income-fall rule; a High rung (CASA balances are swept against overdue)
Persona table	Built; first production run next	Persona and contact-route rules (this deck)

Controls: every change is checked in production against the previous output (distribution and old-vs-new crosstab) before it replaces it; rule tests run on every release; reference workbooks document each axis for GRR.